

TypeScript

Introduction to TypeScript -

1. TypeScript is a programming language, which was developed by Microsoft in the year 2012.
2. TypeScript is a superset of JavaScript which adds more features like static typing, error detection, and code readability.
3. JavaScript is flexible but sometimes it creates bugs because it does not check the type of the data.

// for example -

JavaScript:

```
let age = "20";  
  
console.log(age) // No error
```

TypeScript:

```
let age: Number;  
  
age = "20"  
  
console.log(age) // TypeError
```

Why TypeScript ? -

1. We use typescript because it detects errors at compile time.
2. TypeScript improves code readability and maintainability.
3. It provides strong typing and smart features like auto completion, auto suggestion, error checking.
4. TypeScript makes large projects easy to manage.
5. TypeScript helps testers to write reliable automation scripts.

Why not only JavaScript ? -

1. In JavaScript, most errors are found only when the program runs.
2. Using JavaScript, it is hard to handle big projects.
3. JavaScript does not have type safety (it doesn't check data types).
4. It becomes difficult to maintain large and complex automation frameworks in JavaScript.

// To solve these problems, we use TypeScript.

// TypeScript gives compile-time checking and type safety, so we can find errors early and manage projects easily.

History of TypeScript -

1. TypeScript was introduced in 2012, created by Microsoft and designed by Anders Hejlberg.
2. The reason for the creation of TypeScript was that JavaScript is used for building large web applications but it had problems with

that, so to overcome those problems TypeScript was introduced which was more powerful, safer and maintainable as compared to JavaScript.

TypeScript Growth -

TypeScript become more popular because :

1. It supports the latest JavaScript features.
2. It makes code easy to read and understand.
3. It helps to write better quality code.
4. It helps testers write clean and readable automation scripts.
5. TypeScript works with frameworks like Angular, React, and Node.js. // When Angular adopted TypeScript, it became more popular.

Version of TypeScript -

// TypeScript updates its version regularly.

- 1. version 1:** The first version of TypeScript was released in the year 2012, which was the basic static typing.
- 2. version 2:** In this version , there was better type checking and optional parameters.
- 3. version 3:** In this version, they improved performance and introduced advanced types.

4. version 4: in this version, they introduced faster compilation and powerful type features.

5. version 5: In this version, they enhanced typescript performance and also supported the latest features of JavaScript.

// Currently we are using version 5.9 series.